

Name: _____

1. In Chapter 4, we delved into the production model. Consider a firm with a production function:

$$Y = AK^{\alpha}L^{1-\alpha}$$

This firm rents capital at a rate of r , and the wage rate is w .

- (a) Express the profit maximization problem.
- (b) Calculate the Marginal Product of Capital (MPK) and Marginal Product of Labor (MPL). Show the optimal rule to labor and capital hiring.
- (c) Determine the appropriate course of action for the firm in the following situations:
 - (i) When the marginal product of capital exceeds the rental price of capital.
 - (ii) When the marginal product of labor is less than the wage.

Solution:

- (a) The firm's objective is to maximize profits, calculated as revenues minus costs:

$$\pi = \text{REV} - \text{COST} = F(K, L) - (rK + wL).$$

The firm determines the quantities of capital, K , and labor, L , to employ, taking into account the prevailing market rental rate for capital, r , and the wage rate, w .

- (b) The $MPK = \alpha \frac{Y}{K} = \alpha AK^{\alpha-1}L^{1-\alpha}$ and $MPL = (1 - \alpha) \frac{Y}{L} = (1 - \alpha)AK^{\alpha}L^{-\alpha}$

At optimal, the firm would choose $MPK = r$ and $MPL = w$, which

$$\alpha AK^{\alpha-1}L^{1-\alpha} = r$$

$$(1 - \alpha)AK^{\alpha}L^{-\alpha} = w$$

- (c) (i) If MPK is greater than r , the firm would increase its capital investment since it's producing more output than what it costs to use capital.
- (ii) if MPL is less than w , the firm should reduce its labor input, as the cost of employing a certain number of laborers exceeds the additional output they contribute to.

2. This question pertains to the Solow growth model discussed in Chapter 5. Mount Joy, a closed economy without government spending, features a production function:

$$Y_t = AK_t^{1/3}L_t^{2/3}$$

Here, Y_t represents the output in period t , K_t denotes the capital in period t , and L_t stands for the labor employed in period t . Mount Joy's capital accumulation follows the standard function:

$$K_{t+1} = (1 - \delta)K_t + I_t$$

In this equation, δ represents the depreciation rate, and I_t represents the investment in period t .

Mount Joy's population save a proportion $\bar{s}$ of their output for investment each period. The labor supply is inelastic, with the economy consistently providing $\bar{L}$ units of labor at any given wage.

- (a) Find the steady-state levels of *capital*, *output* and *output per worker* in the Solow model.
- (b) If the initial K_0 is above its steady state value, show and explain the transition dynamics with clear label for the figure and for the movement.

Solution:

(a) We can rewrite the capital accumulation function as

$$\Delta K_{t+1} = I_t - \delta K_t = \bar{s}Y - \delta K_t = \bar{s}AL_t^{2/3}K_t^{1/3} - \delta K_t$$

In steady state, $\Delta K_{t+1} = 0$

$$0 = \bar{s}AL_t^{2/3}K^{1/3} - \delta K$$

Solving for K

$$K^* = \left(\frac{\bar{s}}{\delta} A \right)^{3/2} \bar{L}$$

Plug it into the production function

$$Y^* = A \left(\left(\frac{\bar{s}}{\delta} A \right)^{3/2} \bar{L} \right)^{1/3} \bar{L}^{2/3} = A^{3/2} \left(\frac{\bar{s}}{\delta} \right)^{1/2} \bar{L}$$

The steady state output per capita is then

$$y^* = \frac{Y^*}{L} = A^{3/2} \left(\frac{\bar{s}}{\delta} \right)^{1/2}$$

(b) With $K_0 > K^*$, the net investment is negative $\bar{s}Y - \delta K_t < 0$. This implies the $\Delta K_{t+1} < 0$. With negative net investment, each period, the economy is losing its capital stock and, with less capital available each period, the economy shrinks to the steady state at (K^*, Y^*) over time

